HONG KONG DIPLOMA OF SECONDARY EDUCATION EXAMINATION

MATHEMATICS Compulsory Part PAPER 2

There are 30 questions in Section A and 15 questions in Section B

The diagrams in this paper are not necessarily drawn to scale.
Choose the best answer for each question.

Section A

1. $\dfrac{8^{2n+1}}{4^{3n+1}} =$

A. 1 .

B. 2 .

C. 2^{n} .

D. 2^{-n} .

2. If $\dfrac{a}{x}+\dfrac{b}{y}=3$, then $x=$

A. $\dfrac{ay}{3y-b}$.

B. $\dfrac{ay}{b-3y}$.

C. $\dfrac{by}{3y-a}$.

D. $\dfrac{by}{a-3y}$.

3. $(4x+y)^2-(4x-y)^2=$

A. 0 .

B. $2y^2$.

C. $8xy$.

D. $16xy$.

4. If a , b and c are non-zero constants such that $x(x+3a)+a\equiv x^2+2(bx+c)$, then $a:b:c=$

A. $2:3:1$.

B. $2:3:4$.

C. $3:2:6$.

D. $6:4:3$.

5. Let k be a constant. If $\mathrm{f}(x)=2x^2-5x+k$, then $\mathrm{f}(2)-\mathrm{f}(-2)=$

A. -20 .

B. 0 .

C. 16 .

D. $2k$.

6. Let $f(x) = x^2 - x + 1$. If k is a constant, which of the following must be true?

A. $f(k) = f(-k)$

B. $f(k) = f(1-k)$

C. $f(k+1) = f(k) + f(1)$

D. $f(k-1) = f(k) - f(1)$

7. The solution of $-3x < 6 < 2x$ is

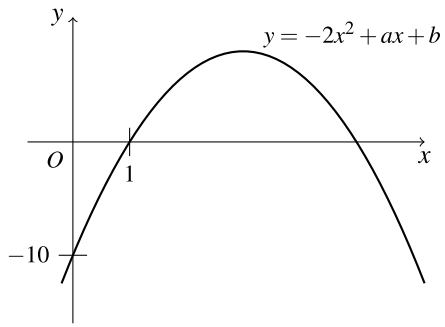
A. $x > -2$.

B. $x > 0$.

C. $x > 3$.

D. $-2 < x < 3$.

8. The figure shows the graph of $y = -2x^2 + ax + b$, where a and b are constants. The equation of the axis of symmetry of the graph is



A. $x = 2$.

B. $x = 3$.

C. $x = 5$.

D. $y = 8$.

9. Let $g(x) = 13 - 5x^2$. If α is a constant, find $g(1 - 3\alpha)$.

A. $8 - 45\alpha^2$

B. $8 + 45\alpha^2$

C. $8 - 30\alpha + 45\alpha^2$

D. $8 + 30\alpha - 45\alpha^2$

10. A car travels at an average speed of 60 km/h for 18 minutes and then the car travels at an average speed of 40 km/h for 27 minutes. The average speed of the car for the whole journey is

A. 48 km/h .

B. 50 km/h .

C. 52 km/h .

D. 54 km/h .

11. It is given that z varies as x^3 and y^2 . When $x = 2$ and $y = 1$, $z = 14$. When $x = 3$ and $y = -2$, $z =$

A. -189 .

B. -126 .

C. 126 .

D. 189 .

12. The actual area of a playground is 900 m^2 . If the area of the playground on a map is 36 cm^2 , then the scale of the map is

A. $1 : 25$.

B. $1 : 50$.

C. $1 : 500$.

D. $1 : 250\,000$.

13. If u varies directly as the square root of v and inversely as w , which of the following are true?

I. u^2 varies directly as v and inversely as the square of w .

II. v varies directly as w and inversely as the square root of u .

III. w varies directly as the square root of v and inversely as u .

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

14. There is a bag of white sugar. The weight of white sugar in the bag is measured as 5 kg correct to the nearest kg . If the bag of white sugar is packed into n packets such that the weight of white sugar in each packet is measured as 10 g correct to the nearest g , find the greatest possible value of n .

A. 429

B. 500

C. 578

D. 579

15. The diameter of a circle is 10 cm . The circle is divided into a major segment and a minor segment by a chord of length 8 cm . Find the area of the major segment correct to the nearest cm^2 .

A. 11 cm^2

B. 23 cm^2

C. 55 cm^2

D. 67 cm^2

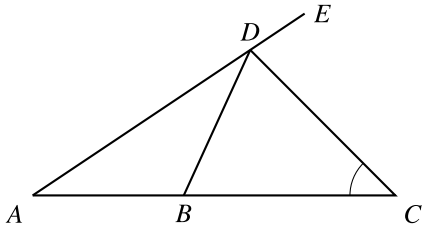
16. In the figure, ABC and ADE are straight lines. It is given that $AB = BD$ and $BC = CD$. If $\angle CDE = 66°$, then $\angle ACD =$

A. $28°$.

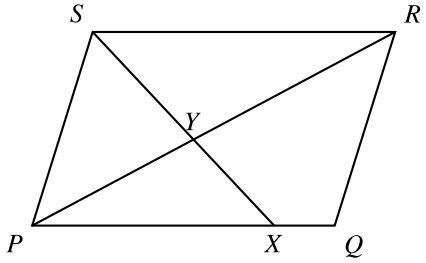
B. $33°$.

C. $36°$.

D. $38°$.



17. In the figure, $PQRS$ is a parallelogram. Let X be a point lying on PQ. Denote the point of intersection of PR and SX by Y. If the area of $\triangle PXY$ and the area of the quadrilateral $QRYX$ are 32 cm^2 and 58 cm^2 respectively, then the area of $\triangle RSY$ is



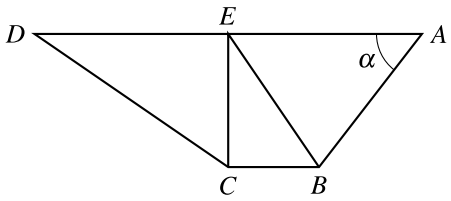
A. 40 cm^2.

B. 50 cm^2.

C. 58 cm^2.

D. 72 cm^2.

18. In the figure, $ABCD$ is a trapezium with $AD \parallel BC$. Let E be the mid-point of AD. It is given that $\angle ABE = \angle BCE = 90^\circ$. Find $\dfrac{CE}{DE}$.



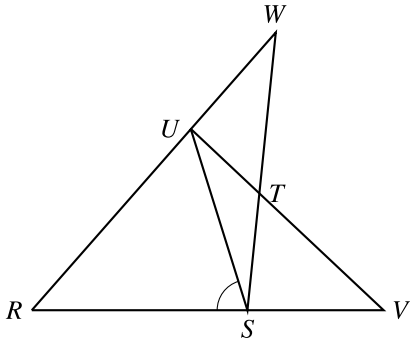
A. $\dfrac{1}{2}$

B. 1

C. $\tan\alpha$

D. $\sin\alpha\cos\alpha$

19. The figure shows the cyclic quadrilateral $RSTU$, where $ST = TU$. RS produced and UT produced meet at the point V while RU produced and ST produced meet at the point W. If $\angle RWS = 32^\circ$ and $\angle RVU = 48^\circ$, then $\angle RSU =$



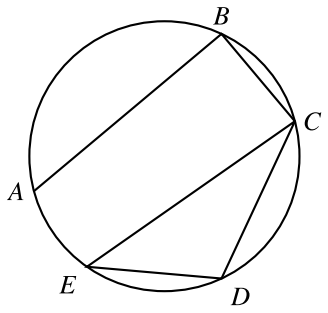
A. 65°.

B. 73°.

C. 80°.

D. 82°.

20. In the figure, $ABCDE$ is a circle. If $AB = 10$ cm, $BC = 5$ cm, $\angle ABC = 90^\circ$ and $\angle CED = 40^\circ$, find CD correct to the nearest cm.



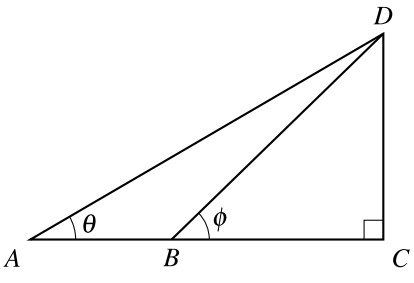
A. 5 cm

B. 6 cm

C. 7 cm

D. 8 cm

21. In the figure, ABC is a straight line. $\dfrac{BC}{AD} =$



A. $\dfrac{\sin\theta}{\tan\phi}$.

B. $\dfrac{\tan\phi}{\sin\theta}$.

C. $\dfrac{\cos\theta}{\tan\phi}$.

D. $\dfrac{\tan\phi}{\cos\theta}$.

22. It is given that $PQRS$ is a rhombus. Let T be the point of intersection of PR and QS. If $\angle QRT = \theta$, then $\dfrac{PQ}{ST} =$

A. $\sin\theta$.

B. $\cos\theta$.

C. $\dfrac{1}{\sin\theta}$.

D. $\dfrac{1}{\cos\theta}$.

23. The coordinates of the point P are $(7,-5)$. P is reflected with respect to the y-axis to the point Q. Q is then rotated clockwise about the origin through 90° to the point R. Find the x-coordinate of R.

A. -7

B. -5

C. 5

D. 7

24. The straight line L is perpendicular to the straight line $9x-5y+45=0$. If the x-intercept of L is -3, then the equation of L is

A. $5x+9y+15=0$.

B. $5x+9y+27=0$.

C. $9x-5y+15=0$.

D. $9x-5y+27=0$.

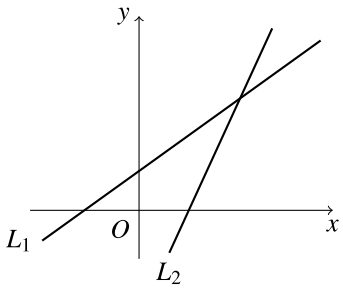
25. In the figure, the equations of the straight lines L_1 and L_2 are $ax+y=b$ and $cx+y=d$ respectively. Which of the following are true?

I. $a<0$

II. $a<c$

III. $b>d$

IV. $ad>bc$



A. I, II and III only

B. I, II and IV only

C. I, III and IV only

D. II, III and IV only

26. The polar coordinates of the points P, Q and R are $(3,160^\circ)$, $(4,280^\circ)$ and $(6,340^\circ)$ respectively. The perpendicular distance from Q to PR is

A. 2.

B. 3.

C. $2\sqrt{3}$.

D. $3\sqrt{3}$.

27. The stem-and-leaf diagram below shows the distribution of the ages of the passengers in a bus.

Stem (tens)	Leaf (units)						
1	h	4	6				
2	3	3	3	4	6	7	7
3	1	2	2	2	6	8	
4	0	k					

If the range of the above distribution is at least 33 , which of the following must be true?

I. $0 \leq h \leq 3$

II. $3 \leq k \leq 9$

III. $3 \leq k - h \leq 5$

A. I only

B. II only

C. I and III only

D. II and III only

28. Consider the following data:

$$19 \quad 10 \quad 12 \quad 12 \quad 13 \quad 13 \quad 14 \quad 15 \quad 16 \quad m \quad n$$

If both the mean and the median of the above data are 14 , which of the following are true?

I. $m \geq 14$

II. $n \leq 16$

III. $m + n = 30$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

29. The mean height of 25 teachers and 140 students is 150 cm . If the mean height of the students is 145 cm , then the mean height of the teachers is

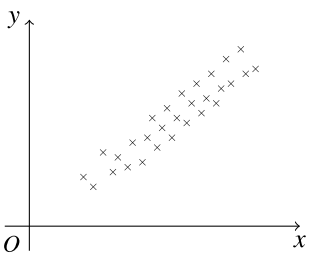
A. 151 cm .

B. 155 cm .

C. 176 cm .

D. 178 cm .

30. The scatter diagram below shows the relation between x and y . Which of the following may represent the relation between x and y ?



A. y increases when x increases.

B. y decreases when x increases.

C. y varies inversely as x^2 .

D. y varies directly as x^{-3} .

Section B

31. The H.C.F. and the L.C.M. of three expressions are ab^2 and $4a^4b^5c^6$ respectively. If the first expression and the second expression are $2a^2b^4c$ and $4a^4b^2c^6$ respectively, then the third expression is

A. ab^2 .

B. ab^5 .

C. $2ab^2c$.

D. $2ab^5c$.

32. The L.C.M. of a^4b^2c , a^3b^4c and $a^2b^5c^2$ is

A. a^2b^2c .

B. $a^2b^2c^2$.

C. a^4b^5c .

D. $a^4b^5c^2$.

33. If $\dfrac{3}{3\log x-2}+7=\dfrac{2}{2\log x+1}$, then $\log\dfrac{1}{x}=$

A. -3 or 2 .

B. -2 or 3 .

C. $\dfrac{-1}{3}$ or $\dfrac{1}{2}$.

D. $\dfrac{-1}{2}$ or $\dfrac{1}{3}$.

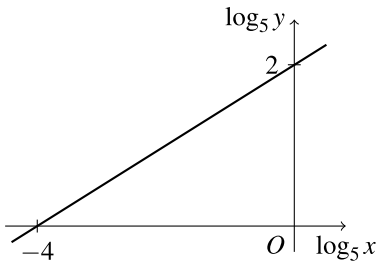
34. The graph in the figure shows the linear relation between $\log_5 x$ and $\log_5 y$. Which of the following must be true?

A. $xy^2=625$

B. $x^2y=625$

C. $\dfrac{y^2}{x}=625$

D. $\dfrac{y}{x^2}=625$



35. If k and $\dfrac{5}{2-i}+ki$ are real numbers, then $k=$

A. -2 .

B. -1 .

C. 1 .

D. 2 .

36. The figure shows a shaded region (including the boundary). If (h,k) is a point lying in the shaded region, which of the following are true?

I. $k \geq 3$

II. $h-k \geq -3$

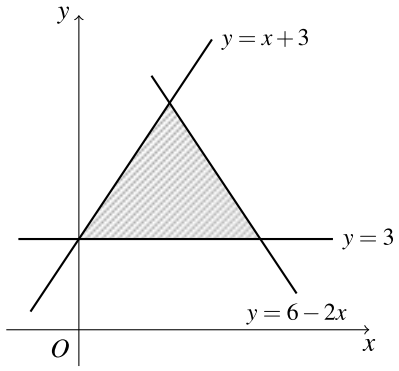
III. $2h+k \leq 6$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III



37. The sum of the 2nd term and the 5th term of a geometric sequence is 9 while the sum of the 7th term and the 10th term of the sequence is 288 . Find the 20th term of the sequence.

A. 65536

B. 131072

C. 262144

D. 524288

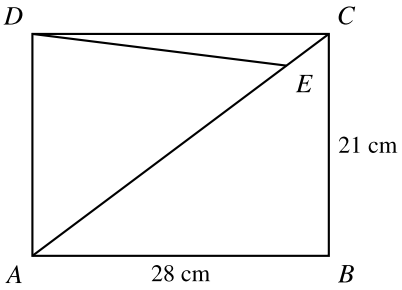
38. In the figure, $ABCD$ is a rectangle. If E is a point lying on AC such that $AE = 30$ cm , then $DE =$

A. $3\sqrt{65}$ cm .

B. $5\sqrt{29}$ cm .

C. $\sqrt{641}$ cm .

D. $\sqrt{697}$ cm .



39. The base of the right pyramid $VABCD$ is the square $ABCD$. Let θ be the angle between $\triangle ABV$ and $\triangle BCV$. If $AB : AV = 5 : 4$, then $\cos\theta =$

A. $\dfrac{-25}{39}$.

B. $\dfrac{-17}{33}$.

C. $\dfrac{-9}{16}$.

D. 0 .

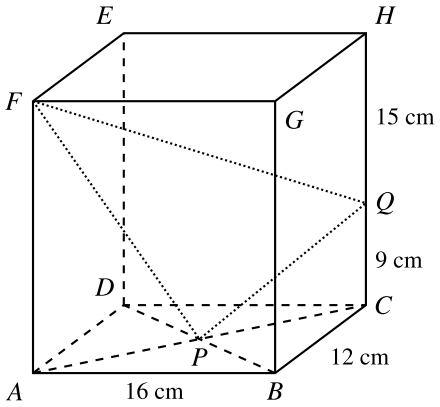
40. In the figure, $ABCDEFGH$ is a rectangular block. AC and BD intersect at P . Q is a point lying on CH such that $CQ = 9$ cm and $QH = 15$ cm . Find $\sin\angle PFQ$.

A. $\dfrac{33}{65}$

B. $\dfrac{56}{65}$

C. $\dfrac{13}{5\sqrt{181}}$

D. $\dfrac{58}{13\sqrt{181}}$



41. The equations of the straight lines L_1 and L_2 are $3x-4y+k=0$ and $4x+3y-k=0$ respectively, where k is a positive constant. It is given that L_1 cuts the x-axis at the point P. Denote the point of intersection of L_1 and L_2 by Q. If R is a point lying on L_2 such that the in-centre of $\triangle PQR$ lies on the x-axis, then the x-coordinate of R is

A. $-7k$.

B. $-k$.

C. k.

D. $7k$.

42. A drama club is formed by 12 boys and 8 girls. If a team of 5 students is selected from the club to participate in a competition and the team consists of at least one girl, how many different teams can be formed?

A. 3960

B. 14712

C. 15448

D. 15504

43. When Teresa throws a dart, the probability that she hits the target is 0.7. If Teresa throws the dart 4 times, find the probability that she hits the target at most 3 times.

A. 0.0081

B. 0.2401

C. 0.7599

D. 0.9919

44. The standard score of Tom in a Mathematics examination is -2. If the score of Tom in the Mathematics examination is 33 marks and the mean of the scores of the Mathematics examination is 45 marks, then the standard deviation of the scores of the Mathematics examination is

A. 3 marks.

B. 6 marks.

C. 12 marks.

D. 36 marks.

45. Let m_1, r_1 and v_1 be the mean, the range and the variance of a group of numbers $\{x_1, x_2, x_3, \ldots, x_{100}\}$ respectively. If m_2, r_2 and v_2 are the mean, the range and the variance of the group of numbers $\{x_1, x_2, x_3, \ldots, x_{100}, m_1\}$ respectively, which of the following must be true?

I. $m_1 = m_2$

II. $r_1 = r_2$

III. $v_1 = v_2$

A. I and II only

B. I and III only

C. II and III only

D. I, II and III

END OF PAPER

ANSWER KEY

1	**B**	16	**A**	31	**B**
2	**A**	17	**B**	32	**D**
3	**D**	18	**D**	33	**D**
4	**A**	19	**B**	34	**C**
5	**A**	20	**C**	35	**B**
6	**B**	21	**A**	36	**D**
7	**C**	22	**C**	37	**C**
8	**B**	23	**B**	38	**A**
9	**D**	24	**A**	39	**A**
10	**A**	25	**C**	40	**A**
11	**D**	26	**C**	41	**C**
12	**C**	27	**B**	42	**B**
13	**B**	28	**D**	43	**C**
14	**C**	29	**D**	44	**B**
15	**D**	30	**A**	45	**A**